Sheet 11 Mon 16 Nov. 2026 *Collinearity, parallelism, intersection of lines*

1. Warm-up · Quick Does the point $(3;5)$ lie on the line $y = 2x - 1$?
2. Core · Method Relative positions of $d:\ y = 3x + 1$ with $d':\ y = 3x - 2$, then with $d'':\ y = -x + 1$.
3. Core + · Method Show in two ways that $A(1;4)$, $B(2;7)$ and $C(4;13)$ are collinear.
4. Review · Quick On a map with scale $\dfrac{1}{50\,000}$, two towns are 4 cm apart. What is the real distance?
5. Thread · Method Programme: choose a number x; add 3; multiply by 2; subtract 6. Calculate the result for $x = 5$, then write an expression in x and simplify it. What do you notice?

Sheet 12 Tue 17 Nov. 2026 *Intersection of two lines; Python collinearity function*

1. Warm-up · Quick Solve $2x + 1 = -x + 7$.
2. Core · Method Coordinates of the intersection of $y = 2x + 1$ and $y = -x + 7$.
3. Core + · Code reading What do `alignes(0,0,1,2,2,4)` and `alignes(0,0,1,2,2,5)` return?

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == 0
```

4. Review · Expand Expand and simplify $3(x - 2)$.
5. Thread · Prove Show that the sum of three consecutive integers is a multiple of 3.

Real numbers and intervals

Sheet 13 Thu 19 Nov. 2026 *Sets of numbers: $\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$*

1. Warm-up · Quick Give the decimal form of $\dfrac{3}{8}$ and an approximation of $\dfrac{5}{6}$ to within 10^{-3}.
2. Core · Method Give the smallest set ($\mathbb{N}, \mathbb{Z}, \mathbb{D}, \mathbb{Q}, \mathbb{R}$) containing each of these numbers: -4; 0.25; $\dfrac{1}{3}$; $\sqrt{2}$; $\dfrac{6}{3}$.
3. Core + · True/false True or false? Justify. The number $\dfrac{1}{3}$ is a decimal number.
4. Review · Quick Calculate $(x + 1)(x - 3)$ for $x = 2$.
5. Thread · Quick How many faces, vertices and edges does a right triangular prism have?

Sheet 14 Mon 23 Nov. 2026 *Decimal bounds and rounding*

1. Warm-up · Quick Round 3.14159 to the nearest 10^{-2} and then to the nearest 10^{-3}.
2. Core · Method Bound $\sqrt{2}$ between two decimals that differ by 10^{-2}.
3. Core + · Code reading What is the final value of `k`? What can be deduced about $\sqrt{2}$?

```
k = 10
while (k/10)**2 < 2:
    k = k + 1
```

4. Review · Quick 5 kg of rice cost 12 500 Ar. How much do 8 kg cost?
5. Thread · True/false True or false? Justify. A quadrilateral whose diagonals have equal length is a rectangle.